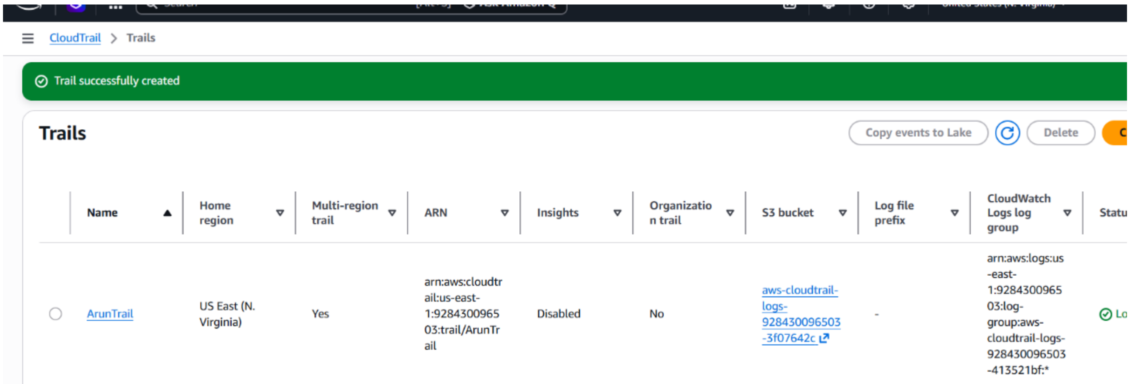
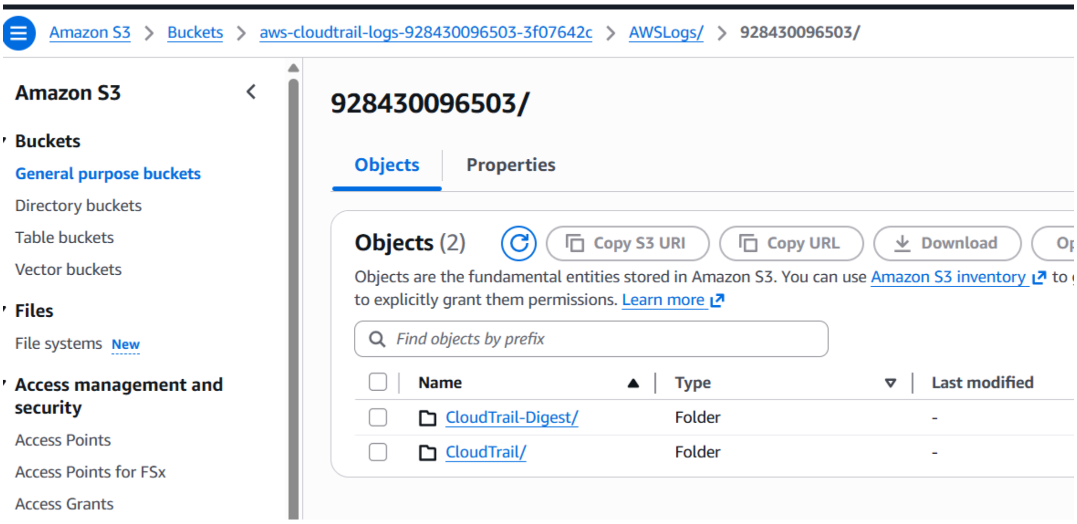
Task on CloudTrail & CloudWatch

1. **Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.**

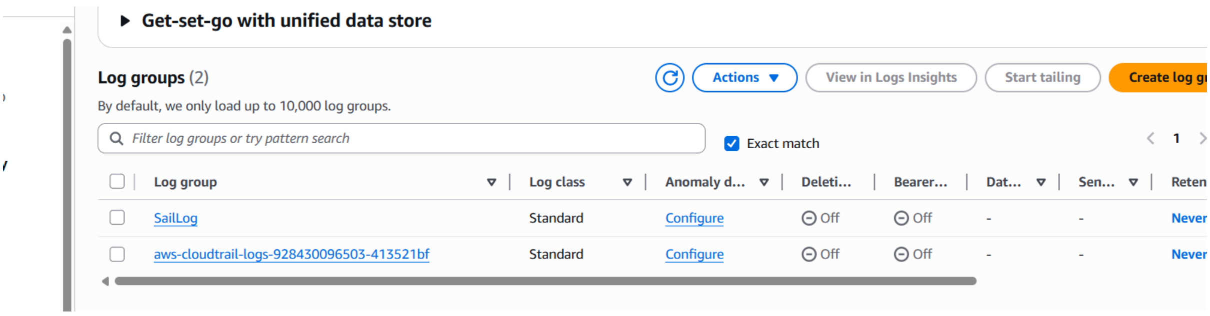
- **CloudTrail Created**

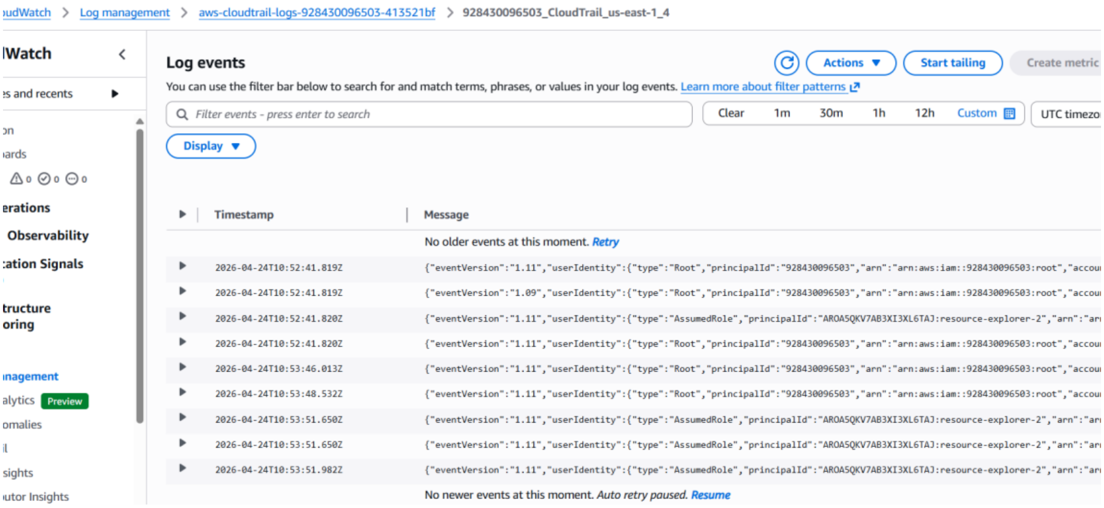


- **Logs Stored in S3**

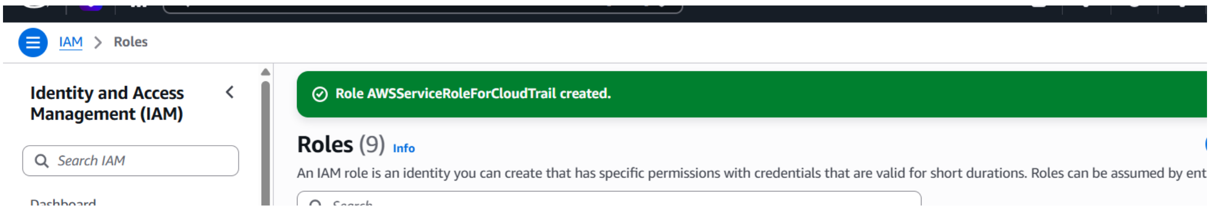


- **CloudWatch Logs**



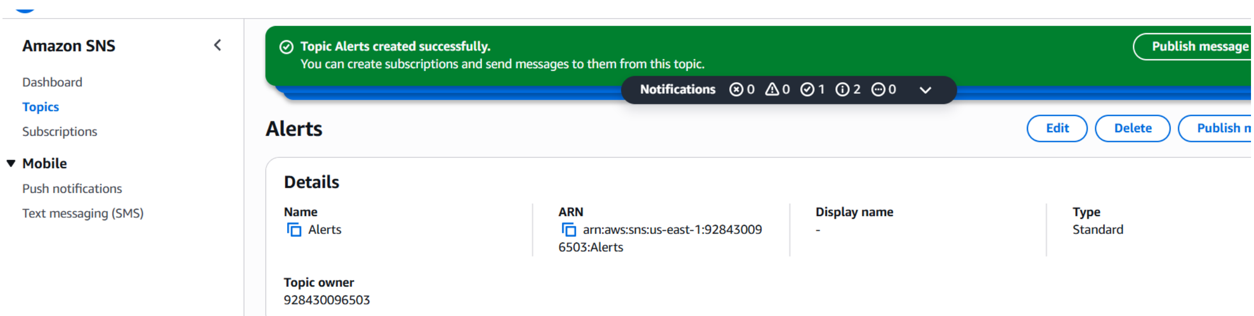


- **IAM Role created**

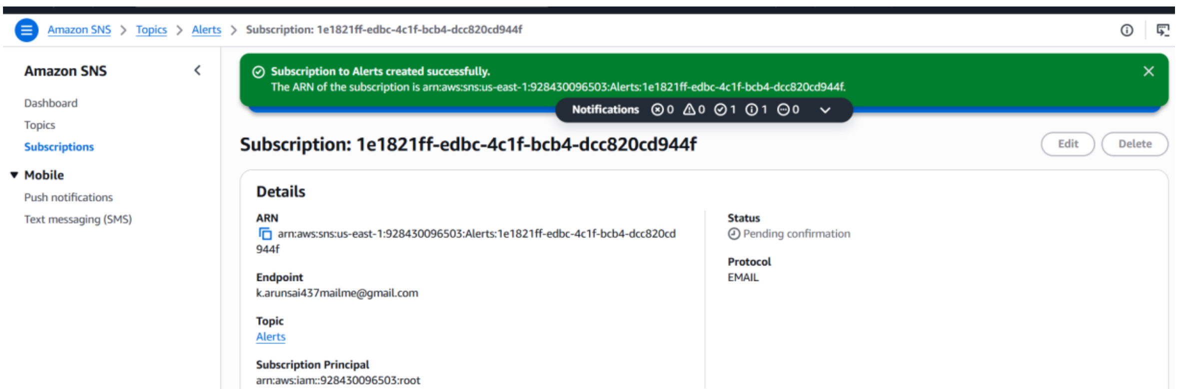


2. **Enable SNS for CloudTrail to send alerts via email.**

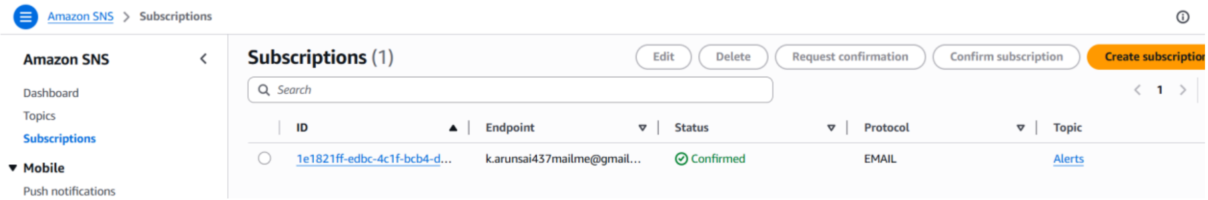
- **Created Alert in SNS**



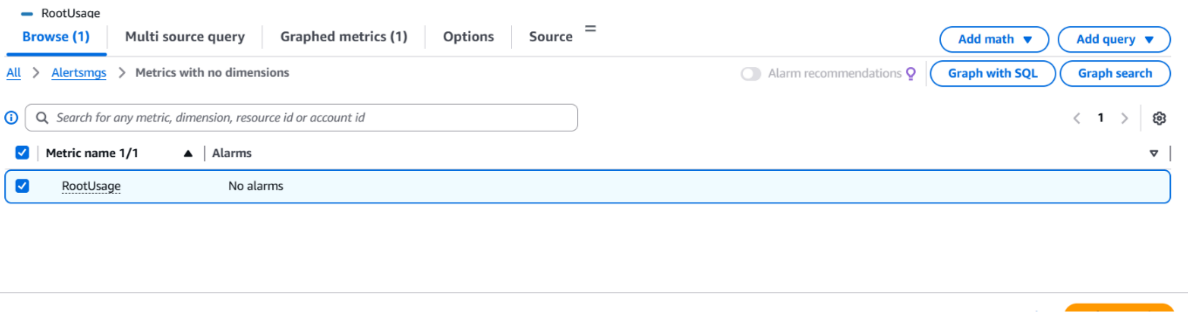
- **Created Protocol Email in SNS**



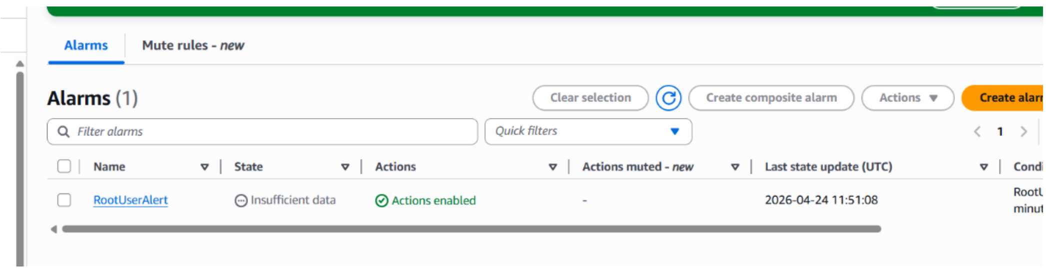
- **Mail Subscription done**



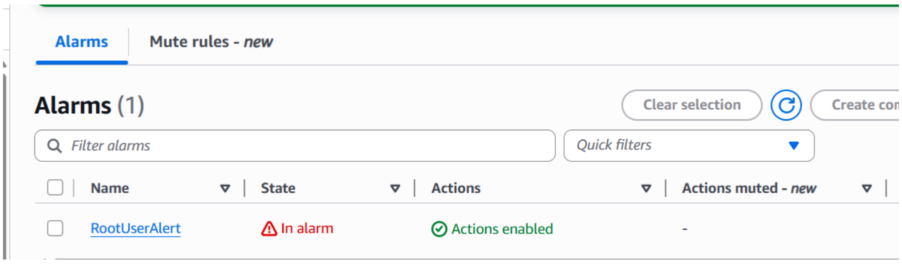
- **Created Metric in CloudWatch Log Groups**

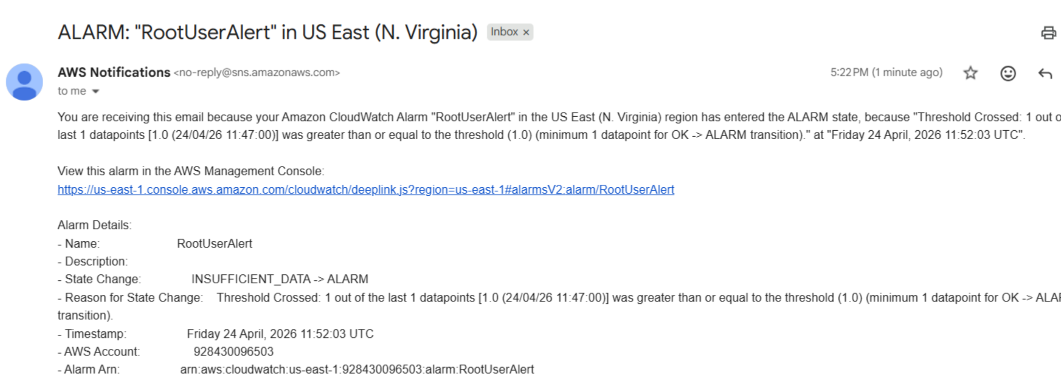


- **Created Alarm**



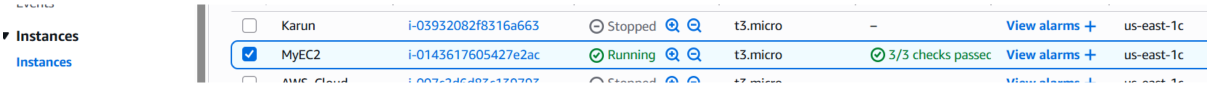
- **Check in mail the alert alarm**



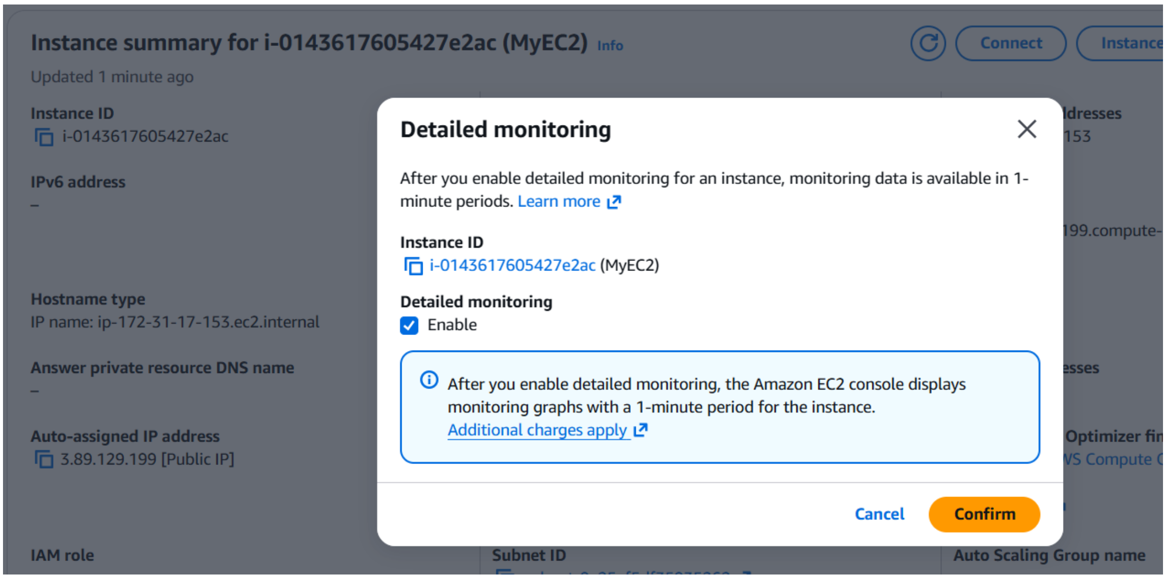


3. **Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.**

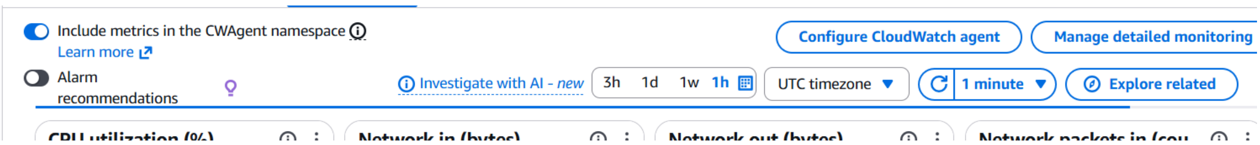
- **Run any instance default**



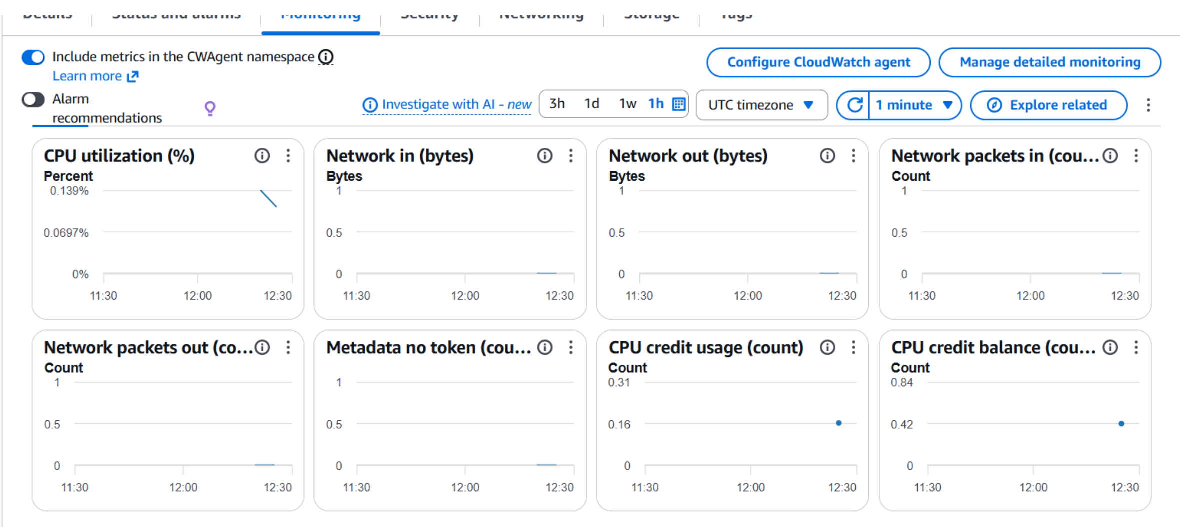
- **In that open Actions and enable the monitoring**



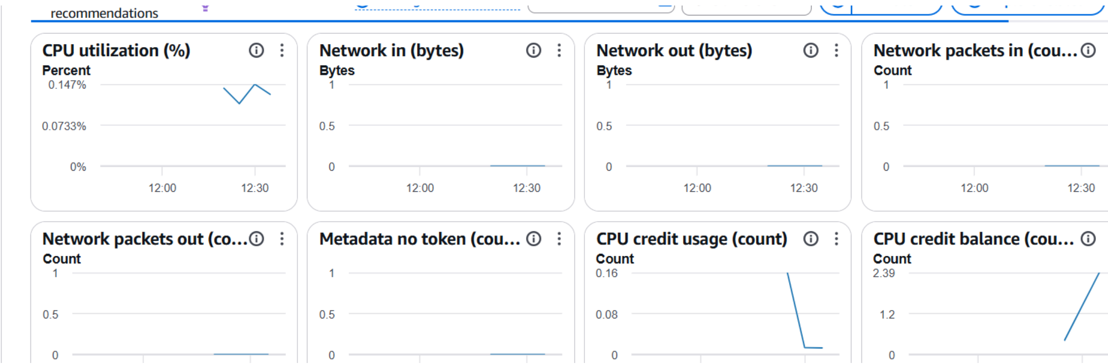
- **Change to 1 Minute for interval metrics**



- **Before it was zero**

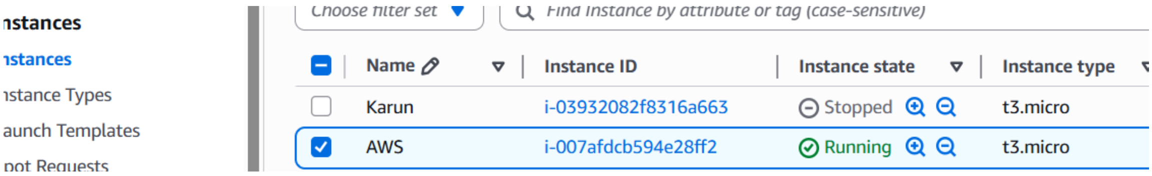


- **After Every 1 Minute the data is increasing Better accuracy for alarms**

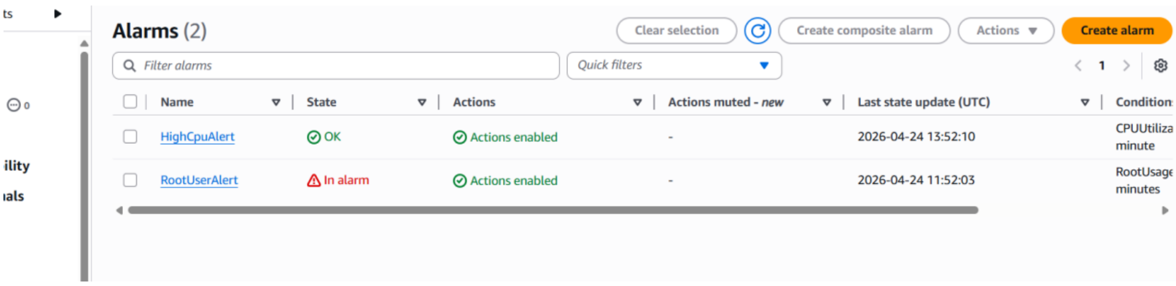


4. **Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.**

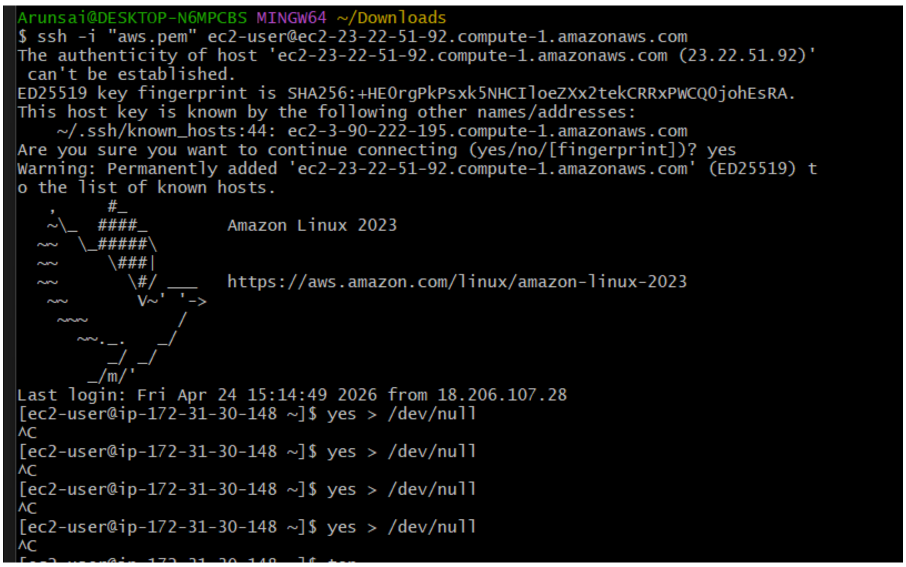
- **RUN any Ec2 in AWS and Connect with pem in gitbash**



- **Create an alert message in alarm the and set more than 70 CPU utilization**

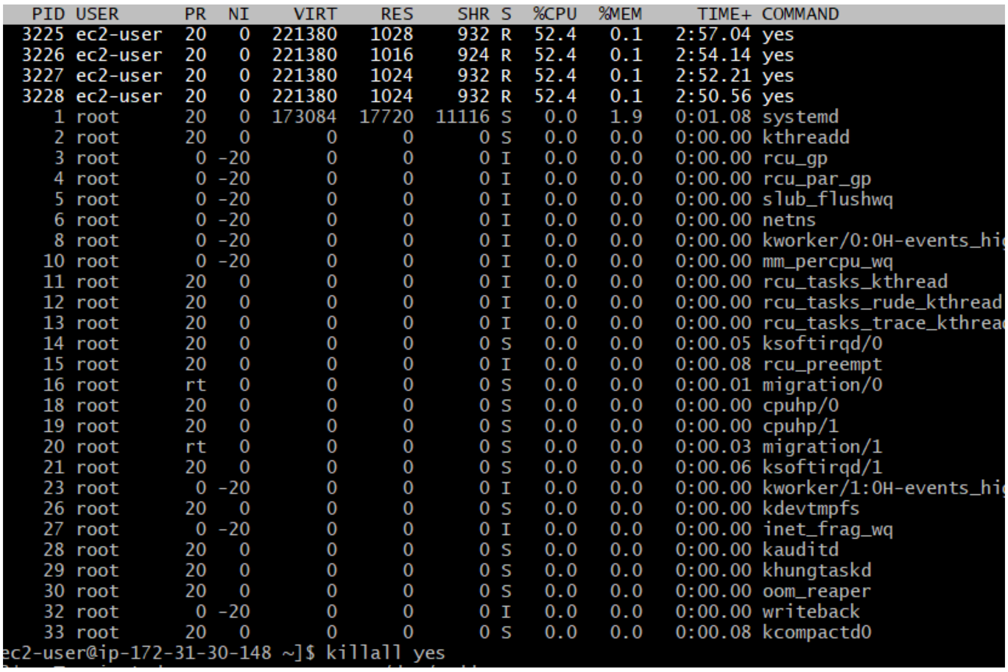


- **Run the EC2 and use the command for increase CPU Utilization**



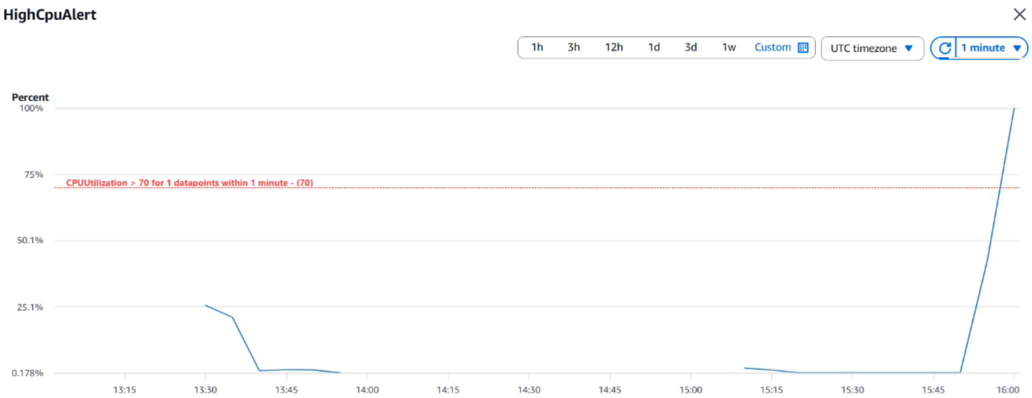
```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ec2-user@ec2-23-22-51-92.compute-1.amazonaws.com
The authenticity of host 'ec2-23-22-51-92.compute-1.amazonaws.com (23.22.51.92)'
 can't be established.
ED25519 key fingerprint is SHA256:+HEOrgPkPsxk5NHCIloeZXx2tekCRRxPWCQOjohEsRA.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:44: ec2-3-90-222-195.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-23-22-51-92.compute-1.amazonaws.com' (ED25519) t
o the list of known hosts.
                    Amazon Linux 2023
                    https://aws.amazon.com/linux/amazon-linux-2023
Last login: Fri Apr 24 15:14:49 2026 from 18.206.107.28
[ec2-user@ip-172-31-30-148 ~]$ yes > /dev/null
^C
[ec2-user@ip-172-31-30-148 ~]$ yes > /dev/null
^C
[ec2-user@ip-172-31-30-148 ~]$ yes > /dev/null
^C
[ec2-user@ip-172-31-30-148 ~]$ yes > /dev/null
^C
```

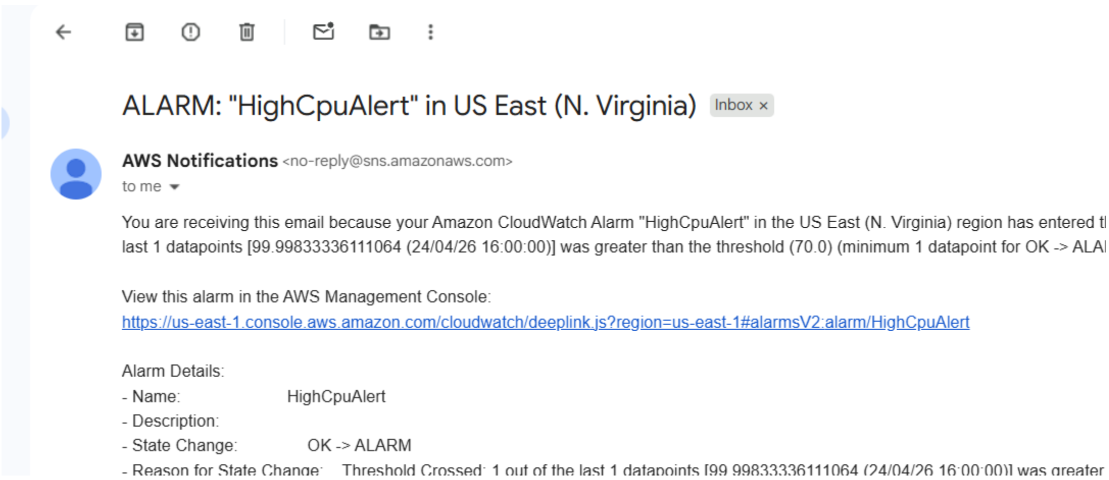
- **Once the CPU utilization increase to 100 kill the process**



```
  PID USER       PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 3225 ec2-user   20   0  221380   1028    932 R  52.4   0.1   2:57.04 yes
 3226 ec2-user   20   0  221380   1016    924 R  52.4   0.1   2:54.14 yes
 3227 ec2-user   20   0  221380   1024    932 R  52.4   0.1   2:52.21 yes
 3228 ec2-user   20   0  221380   1024    932 R  52.4   0.1   2:50.56 yes
    1 root       20   0  173084  17720  11116 S   0.0   1.9   0:01.08 systemd
    2 root       20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
    6 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
    8 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_hi
   10 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
   11 root       20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   12 root       20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   13 root       20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthrea
   14 root       20   0       0      0      0 S   0.0   0.0   0:00.05 ksoftirqd/0
   15 root       20   0       0      0      0 I   0.0   0.0   0:00.08 rcu_preempt
   16 root       rt   0       0      0      0 S   0.0   0.0   0:00.01 migration/0
   18 root       20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
   19 root       20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
   20 root       rt   0       0      0      0 S   0.0   0.0   0:00.03 migration/1
   21 root       20   0       0      0      0 S   0.0   0.0   0:00.06 ksoftirqd/1
   23 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_hi
   26 root       20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
   27 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
   28 root       20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
   29 root       20   0       0      0      0 S   0.0   0.0   0:00.00 khungtaskd
   30 root       20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
   32 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 writeback
   33 root       20   0       0      0      0 S   0.0   0.0   0:00.08 kcompactd0
ec2-user@ip-172-31-30-148 ~]$ killall yes
```

- **We can check in CloudWatch and also in mail**





ALARM: "HighCpuAlert" in US East (N. Virginia) Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com>
to me

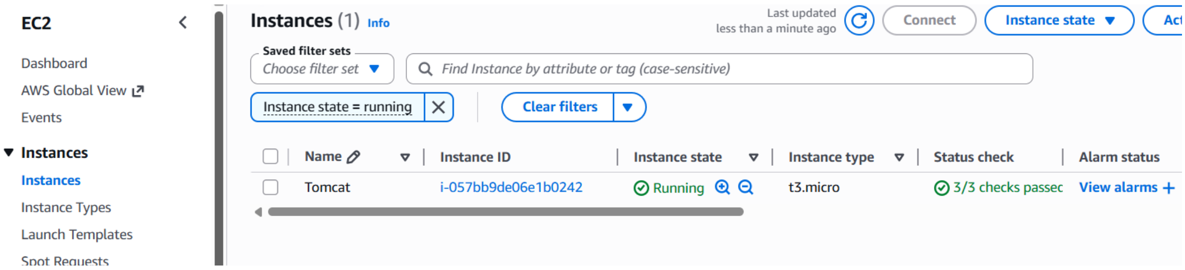
You are receiving this email because your Amazon CloudWatch Alarm "HighCpuAlert" in the US East (N. Virginia) region has entered t
last 1 datapoints [99.99833336111064 (24/04/26 16:00:00)] was greater than the threshold (70.0) (minimum 1 datapoint for OK -> ALA

View this alarm in the AWS Management Console:
https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/HighCpuAlert

Alarm Details:
- Name: HighCpuAlert
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [99.99833336111064 (24/04/26 16:00:00)] was greater

5. **Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.**

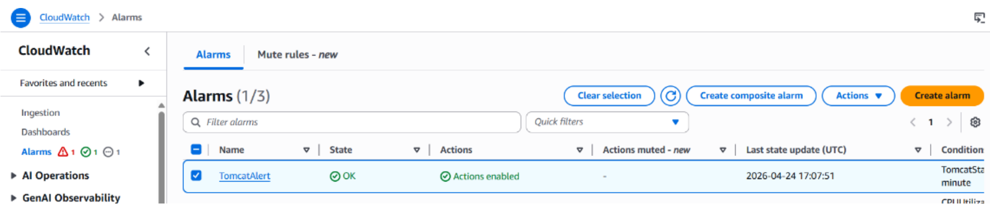
- **New instance created and run**



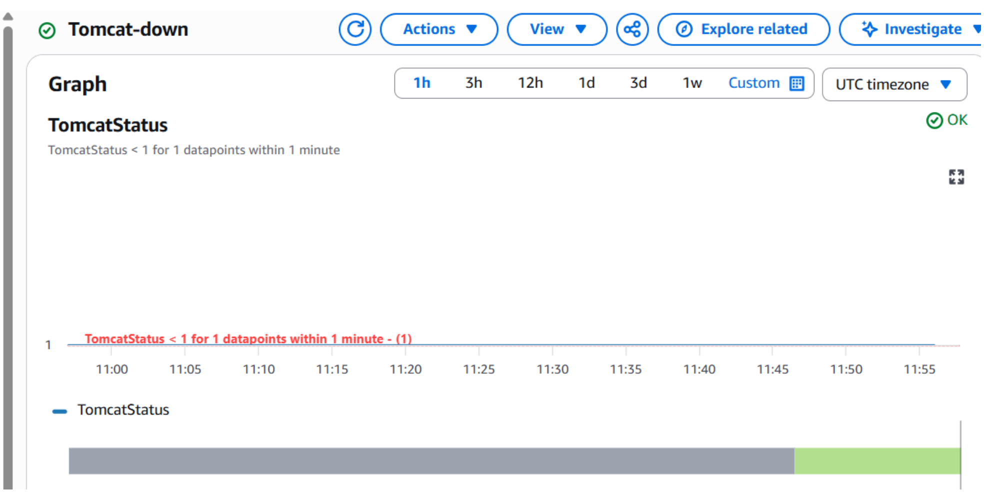
- **Installed tomcat Apache and Run it**

```
fi
[ec2-user@ip-172-31-28-57 ~]$ ./tomcat-check.sh
[ec2-user@ip-172-31-28-57 ~]$ sudo su
[root@ip-172-31-28-57 ec2-user]# sudo pkill -f tomcat
[root@ip-172-31-28-57 ec2-user]# cd /opt/tomcat/bin
[root@ip-172-31-28-57 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

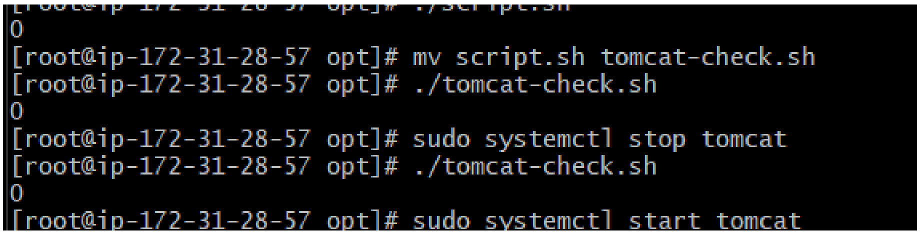
- **Create an alarm for tomcat**



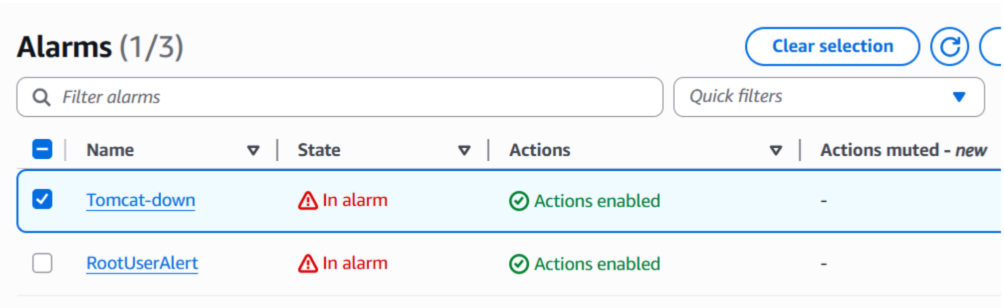
- **For alarm check in metric and value given low and attached to SNS**



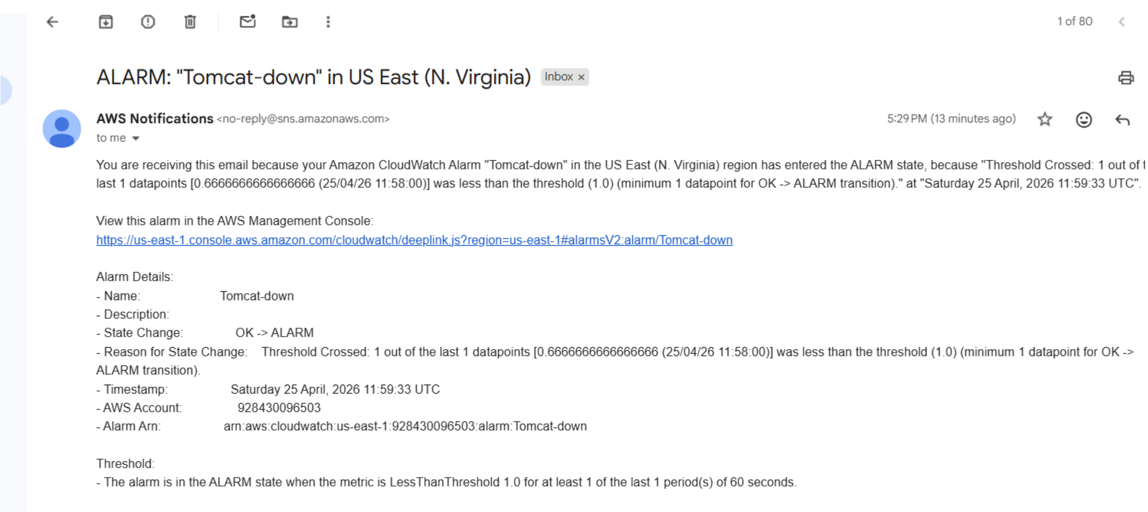
- **Stop the tomcat server and check the alert alarm**



- **We can see the alarm in Alarms in CloudWatch of tomcat**

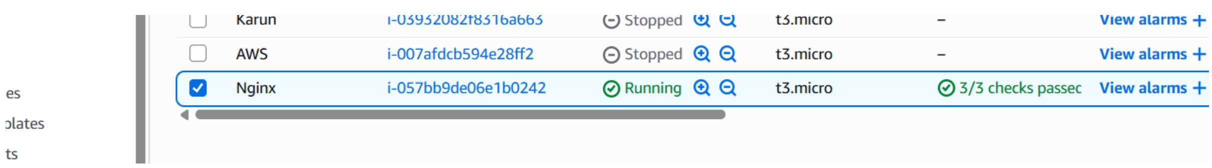


- **We can same thing in mail via SNS we get**



6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

- **Created new instance Nginx in EC2 an run it**

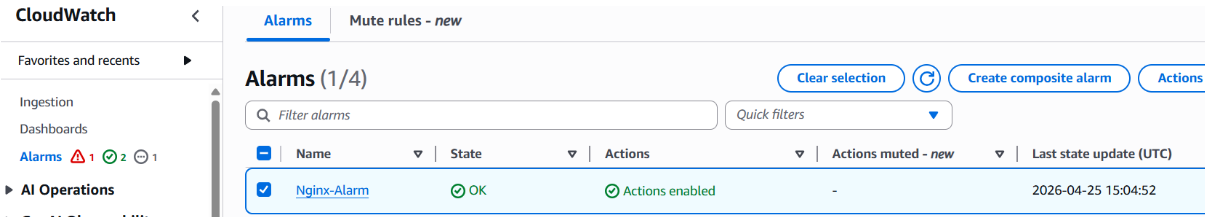


- **Installed Nginx and run it**

```
[ec2-user@ip-172-31-28-57 ~]$ systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: di>
     Active: active (running) since Sat 2026-04-25 14:57:56 UTC; 10s ago
    Process: 3413 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, stat>
    Process: 3414 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCES>
    Process: 3415 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 3416 (nginx)
      Tasks: 3 (limit: 1014)
     Memory: 3.4M
        CPU: 57ms
     CGroup: /system.slice/nginx.service
             ├─3416 "nginx: master process /usr/sbin/nginx"
             ├─3417 "nginx: worker process"
             └─3418 "nginx: worker process"

Apr 25 14:57:56 ip-172-31-28-57.ec2.internal systemd[1]: Starting nginx.service>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal nginx[3414]: nginx: the configurat>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal nginx[3414]: nginx: configuration >
```

- **Created an alarm in cloud watch and threshold value is lower than 1**

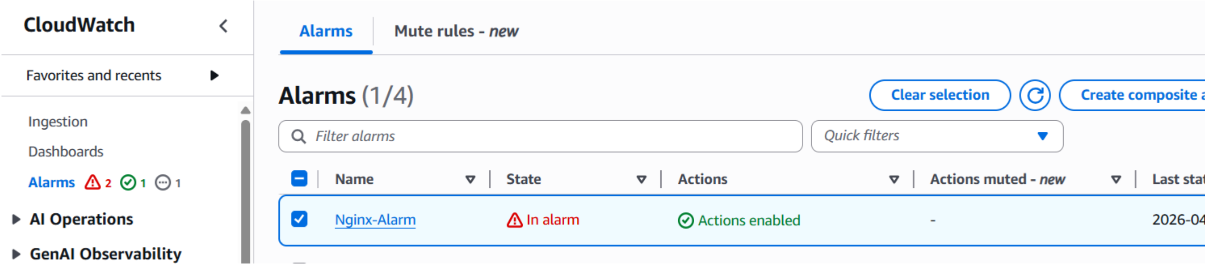


- **Stop the nginx and check the alarm and mail**

```
[ec2-user@ip-172-31-28-57 ~]$ sudo systemctl stop nginx
[ec2-user@ip-172-31-28-57 ~]$ sudo systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: di>
     Active: inactive (dead) since Sat 2026-04-25 15:05:52 UTC; 17s ago
   Duration: 7min 55.334s
   Main PID: 3416 (code=exited, status=0/SUCCESS)
        CPU: 59ms

Apr 25 14:57:56 ip-172-31-28-57.ec2.internal systemd[1]: Starting nginx.service>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal nginx[3414]: nginx: the configurat>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal nginx[3414]: nginx: configuration >
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal systemd[1]: Started nginx.service >
Apr 25 15:05:52 ip-172-31-28-57.ec2.internal systemd[1]: Stopping nginx.service>
Apr 25 15:05:52 ip-172-31-28-57.ec2.internal systemd[1]: nginx.service: Deactiv>
Apr 25 15:05:52 ip-172-31-28-57.ec2.internal systemd[1]: Stopped nginx.service >
lines 1-14/14 (END)
```

- **Check the alarm got the alarm on and mail also**



- **Got mail the alarm for Nginx**